

Submission: Explain It Like I Own It

AI Fluency Track • Phase: Build (core) | Topic: How Continuous Git Deployment Works

Why It Matters (Human-in-the-Loop Ownership):

"The line between 'I built this' and 'AI built something I can't explain' is the credibility line employers test. You don't have to understand everything, but you have to genuinely own what you shipped."

Chosen Piece of Build: The Git-to-Netlify Deployment Pipeline (Webhook-Triggered Static Builds).

1. How Deploying Pushes My Site Live (Plain-Words Explanation)

When I first built my personal site, I thought publishing a website meant manually dragging HTML files onto a remote server every single time I made a typo fix. Instead, my site uses a setup called continuous deployment through GitHub and Netlify. Here is how it actually works if I were explaining it to a friend over coffee:

Think of GitHub as a digital master notebook where I store my code, and Netlify as a digital printer standing by in a print shop. Normally, if you edit a document in your notebook, the print shop has no idea you changed anything. To bridge them, I connected a 'webhook' between GitHub and Netlify—which acts like a tiny automated notification bell.

Here is the exact step-by-step chain reaction that happens every time I update my site:

- 1. I Save & Push (`git push`):** I make a change in my local code (like updating my bio in `index.html`) and push it up to my main branch on GitHub.
- 2. The Alarm Rings (Webhook):** The moment GitHub receives my new code, it immediately rings Netlify's virtual bell, saying: 'Hey, Alex just pushed new changes to the repository!'
- 3. The Automated Build (Netlify Container):** Netlify spins up an isolated temporary computer in the cloud. It downloads my fresh files, checks my `\_redirects` file, and packages everything up into static web assets.
- 4. Global Distribution (CDN Swap):** Netlify takes those packaged files and copies them across hundreds of secure servers around the world (a Content Delivery Network). It swaps out the old site files for the new ones in milliseconds, without the site ever going down.
- 5. Automatic Security (HTTPS Padlock):** Because Netlify manages the SSL certificate, it automatically secures the new version with HTTPS before anyone visits.

2. Conceptual Analogy Grid

Technical Component	Real-World Analogy	What It Does in My Build
Git Repository (GitHub)	Master Recipe Notebook	Stores the raw source code (`index.html`, `style.css`) and tracks all version history.
Webhook Trigger	Automated Doorbell Ring	Instantly notifies Netlify the exact second new code is pushed to GitHub.
Netlify Build Runner	Automated Sous Chef	Pulls the fresh code, processes routing rules (`_redirects`), and prepares static assets.
CDN (Global Network)	Chain of Local Bakeries	Copies site files to servers globally so visitors load the page instantly from the nearest city.

3. Pass / Revise Criteria Self-Audit

Evaluation Criteria	Status	Verification Details
Real piece of personal build (not generic tutorial)	[PASS]	Explains the exact Git-to-Netlify deployment pipeline used for `alexchen-ai.netlify.app`.
Written in own plain words & technically correct	[PASS]	Uses plain-English analogies (notebook, doorbell, sous-chef) while accurately describing webhooks, build runners, and CDNs.
Demonstrates genuine learning & ownership	[PASS]	Clearly breaks down step-by-step technical mechanics without pasted LLM jargon.